A Metric-Order Obstruction for Classical Harmonic Media

A framework-independent statement of the theorems governing when a material substrate can, and cannot, mimic the weak-field metric of general relativity. July 2026.

Abstract

We consider the class of CLASSICAL HARMONIC MEDIA: material substrates whose local properties respond to local state, whose static conditioning fields are harmonic outside localized sources, and in which waves, clocks (bound modes), and rulers (equilibrium structures) are co-material — built from the same substrate properties. We prove four results. (1) An equivalence-principle identity: the locally measured wave speed is invariant under arbitrary conditioning, so no local experiment detects the medium's state. (2) A single consistent PPN-type parameter γ is defined by the response ratios, and $\gamma = 1$ holds iff the responses satisfy one linear relation. (3) The metric-order obstruction: material response to displacement-sourced fields is capped at tidal order ($1/r^3$), one derivative above the potential order ($1/r$) that metric phenomenology requires — while the substrate reproduces the correct TIDAL tensor shape exactly, by harmonicity. (4) Sourcing exhaustion: channels reaching potential range are direction-blind (diffusive) and sign-locked; direction-carrying structure dilutes by solid angle ($1/r^2$). Consequently every classical mechanism of this class yields $\gamma \in [-1, 0]$, explaining a century of mechanical gravity theories that captured Newton and failed light deflection. The results are stated with their premises as an explicit attack surface, including the topological exception and its closures.

1. Definitions and premises

A classical harmonic medium consists of: a substrate with local properties (tension-like T, density-like μ, rigidity-like λ, and structural state such as path tortuosity σ); wave excitations propagating at $c = \sqrt{(T/\mu)}$ along the substrate; bound modes (clocks) whose standing-wave conditions live on the substrate's arc length; and equilibrium structures (rulers) whose extents are set by mode scales $\xi = \sqrt{(\lambda/T)}$. TWO PRIMITIVE PREMISES (an audit shows commonly listed additional premises reduce to these): (P1) local response — local constitutive energy is invariant under rigid translation and therefore depends on deformation gradients rather than absolute displacement (uniform translation is unobservable; spatially varying displacement is physically meaningful; multivalued/topological sectors are physical and treated in Sec. 6); (P2) harmonic statics — outside localized sources, static conditioning potentials satisfy Laplace's equation.

2. The equivalence-principle identity and the single gamma

For conditioning $(\tau, m, l, \sigma) = (dT/T, d\mu/\mu, d\lambda/\lambda, \text{tortuosity})$, the responses are $dc/c = (\tau-m)/2 - \sigma$; $dL/L = (l-\tau)/2 - \sigma$; $d\omega/\omega = \tau - (m+l)/2$. THEOREM (EP identity): $dc/c - dL/L - d\omega/\omega = 0$ identically, for arbitrary conditioning including the structural channel — co-materiality makes the medium's state locally undetectable; this is an equivalence-principle-like statement derived, not assumed. COROLLARY: the bending and ruler-geometry definitions of γ

coincide identically; the medium defines ONE $\gamma\_eff = (dL/L)/(d\omega/\omega)$, and $\gamma = 1$ iff $m = 3\tau - 2l + 2\sigma$-corrections — precisely, $\sigma = l + m/2 - (3/2)\tau$ in the presence of the structural channel.

3. The metric-order obstruction

By (P1), material properties respond to gradients of displacement (strain), not displacement. In three-dimensional elasticity a point FORCE gives displacement ~ $1/r$ (the Kelvin solution) and strain ~ $1/r^2$ — so the radial cap requires a lemma, not an assumption. NO-MONOPOLE LEMMA: a static, isolated defect in equilibrium exerts zero net force on the medium (else it accelerates); the Kelvin amplitude IS the net force and vanishes for every admissible static mass-source. By (P2) the leading admissible displacement is therefore the force-free order u ~ $1/r^2$; hence strain, and every displacement-sourced conditioning, is capped at $1/r^3$ — TIDAL order. Metric phenomenology (redshift, deflection, delay at observed strength and range) requires conditioning at POTENTIAL order, $1/r$. Meanwhile the strain field around a point defect is the Hessian of the harmonic potential, with eigenvalues proportional to $(2, -1, -1)$: the substrate contains the correct tidal tensor shape EXACTLY — one derivative too high. This is the central obstruction: right geometry, wrong order, as a theorem rather than a failure of construction.

4. Sourcing exhaustion at potential range

Structures reaching $1/r$ are diffusive scalars (Laplace fields of absorbed or emitted flux) and are DIRECTION-BLIND: diffusion erases angular memory, so their conditioning is isotropic, and every response channel they drive is SIGN-LOCKED to them (a medium has one such scalar unless its commitments contain a second independent harmonic field). Direction-carrying structure near a source — anchored filaments, ballistic shadows, structural anisotropy, connectivity excess — has a local fraction that is a solid-angle fraction: it dilutes as $1/r^2$, by geometry rather than by modeling choice, and its tension-flux is conserved through spheres (branching topology cannot change the exponent). Consequently: isotropic pure channels give $\gamma = -1/2$ (tension), 0 (density), -1 (rigidity); the structural channel gives $\gamma = -1/2 - \rho$ with $\rho > 0$ for same-scalar sourcing; and the traceless $1/r$ conditioning at locked ratio 1/4 that general relativity requires is unreachable. All classical mechanisms of this class yield $\gamma \in [-1, 0]$.

5. The historical corollary

Newtonian phenomenology is the FIXED POINT of conservation plus three-dimensional geometry: any medium attaching conserved stress or flux to masses obtains the $1/r^2$ force essentially for free — which is why mechanical accounts of Newtonian attraction have been plentiful for three centuries. The same conservation caps those media at tidal or force order and confines their γ to $[-1, 0]$, which is why every mechanical and emitter theory of gravity managed orbits and failed light deflection: the 1919 eclipse excluded a CLASS, and these theorems say which class and why. A mechanism's Newtonian success and its Einsteinian failure can be the same mathematical fact.

6. Scope, the topological exception, and the attack surface

The theorems' premises are their complete attack surface, listed for adversaries. (P1)'s exception is genuine: where the medium supports topological defects, the multivalued part of displacement is physical (holonomy), and such media generically possess a gauge-field sector — in the motivating substrate, that sector is electromagnetism itself (charge as winding). Three closures prevent its recruitment for gravity: topologically NEUTRAL sources yield dipole-and-faster fields, caught by the range results; coupling gravity to the topological charge would violate the universality of free fall, excluded experimentally to $\sim 10^{-13}$; and global constraints contribute zero-modes that cannot source per-mass 1/r fields. OUTSIDE SCOPE, deliberately: quantum-induced dynamics (Sakharov-type loop generation of covariant metric dynamics), non-equilibrium substrates, and media with a second independent harmonic scalar — the last being the exact specification of what a counterexample must contain. Conditions that would overturn these results: a classical channel evading solid-angle dilution; an anti-correlated second harmonic field arising from stated commitments; or a demonstrated 1/r traceless conditioning at locked ratio. The authors know of none, within the presently identified classes.

Revision (July 2026, on external review): hypotheses made explicit; the bound made class-specific

Four precision amendments on specialist critique, each strengthening the statement by narrowing it. (1) The gauge claim now reads in its exact form (translation invariance of the constitutive energy). (2) The radial-order cap now rests on the NO-MONOPOLE LEMMA rather than an implicit force-neutrality assumption; the Kelvin 1/r loophole is real elasticity and is closed by statics, with the lemma reproduced in Section 3 and machine-verified in the corpus (GRV-017). (3) The sourcing-exhaustion result operates under EXPLICIT HYPOTHESES, now part of the theorem: H1 one independent harmonic scalar; H2 local constitutive response; H3 quasi-static transport; H4 no scale-free nonlocal kernel; H5 no independent tensor order parameter; H6 no long-range orientational phase. (4) The $\gamma \in [-1, 0]$ bound is CLASS-SPECIFIC: every mechanism within the defined class of local, harmonic, co-material responses — not all conceivable classical media; engineered media with multiple coupled fields, nonlocal laws, tensor order parameters, or background flows lie outside the class and outside this claim. Correspondingly, quantum-induced dynamics is A leading escape, not the unique one: a second anti-correlated harmonic field (violating H1), nonlocal constitutive laws (H4), non-equilibrium substrates (H3), or irreducible tensor order parameters (H5) are logical alternatives, each a different ontology. These theorems await, and invite, specialist review of exactly these load-bearing steps.

Second revision (July 2026): H1 attacked and defended; the counterexample specification made concrete

A specialist attack proposed violating H1 through the substrate's INTERNAL structure: a two-strand medium's relative (braid) sector as a second, neutral harmonic field with anti-correlated response. The audit produced the INTERNAL-MODE DICHOTOMY: every internal mode is either gapped — hence Yukawa-screened, wrong range — or gapless by a continuous symmetry, hence a phase whose local physical content is its gradient, replaying the no-monopole ladder; and a two-strand medium's internal continuous symmetry group has exactly one generator (the screw mode, coupling slide and rotation), which in the motivating substrate is already the electromagnetic sector and is unsourceable at monopole order by neutral masses. H1 therefore survives, and its counterexample specification sharpens from the abstract 'a second independent harmonic scalar' to the concrete: a substrate whose internal continuous symmetry group possesses AT LEAST TWO generators, with the second neutrally sourceable. This is a well-posed symmetry-group question, and it is the exact door these theorems guard.

Appendix: the internal symmetry theorem (formal), and the emergent-tensor closure

THE THEOREM, made watertight per specialist review. Definition: fixing the centerline, the internal state of a two-strand medium is the azimuth field $\varphi(s)$ of the strand-separation vector (the separation magnitude is gapped: a bound pair has a restoring potential). The internal symmetry group is $G = \mathbb{R}$_slide × SO(2)_rot, acting by (a, α): $\varphi(s) \to \varphi(s - a) + \alpha$; dim G = 2. The helical ground state $\varphi_0(s) = \tau_0 s$ has stabilizer $H = \{\alpha = \tau_0 a\}$ — the screw subgroup, dim 1. THEOREM: the ground-state manifold G/H has dimension one and topology S^1; the internal sector contributes exactly ONE Goldstone field Φ, circle-valued. COROLLARY 1 (allocation is topology): $\pi_1(S^1) = \mathbb{Z}$ quantizes the closed-loop winding of Φ — in the motivating substrate, charge — and Φ's wave dynamics is torsion — light. COROLLARY 2 (angular no-monopole, the rotational twin of the Kelvin closure): a static isolated defect in equilibrium exerts zero net torque on the medium, else it spins up; Φ's monopole amplitude is the sum of net torque and net winding, both zero for neutral static sources, so Φ-sourcing begins at dipole order, $\geq 1/r^2$. Any two-strand medium whose slide and rotation symmetries are broken to a screw by helical order has this structure; the H1 counterexample specification is thereby formal: an internal continuous symmetry group with at least two generators, the second neutrally sourceable.

THE EMERGENT-TENSOR CLOSURE (H5's coarse-grained version). An emergent nematic order Q_{ij} from orientation correlations is closed on every branch, with the closure type labeled honestly per branch. Isotropic vacuum (the class's commitment): Landau-de Gennes gives Q a mass term — gapped, screened; induced alignment is sourced by the anchored population at $1/r^2$. Nematic vacuum: NOT closed by range — Frank elasticity makes director perturbations harmonic, reaching $1/r$, which is precisely why H6 exists — but the order itself is excluded by observed vacuum isotropy (photon-sector anisotropy bounds), by the class's exact emergent Lorentz invariance, and by the absence of any ordering mechanism in an isotropic bath. Near-critical enhancement is the scale-free rescue in tensor costume and is refused on the same audit standard. Emergent tensors either stay gapped or become the long-range orientational phase H6 names; that phase contradicts the vacuum we observe.